

Code

```
import pandas as pd
import numpy as np

Fintrust = pd.read_csv('transactionfintrust.csv')
#fintrust transaction data

Trustfin = pd.read_csv('Customerfintrust.csv')

Fintrust
Transaction_ID Customer_ID Transaction_DateTime Transaction_Type \
0 FT-T000001 FT-C01135 01/01/2026 00:00 Card Purchase
1 FT-T000002 FT-C00428 01/01/2026 00:10 Cash Withdrawal
2 FT-T000003 FT-C00469 01/01/2026 00:21 Transfer
3 FT-T000004 FT-C01015 01/01/2026 00:32 Deposit
4 FT-T000005 FT-C00870 01/01/2026 00:43 Cash Withdrawal
... ..
11995 FT-T011996 FT-C01367 3/31/2026 23:16 Card Purchase
11996 FT-T011997 FT-C00395 3/31/2026 23:27 Airtime/Data
11997 FT-T011998 FT-C00484 3/31/2026 23:38 Card Purchase
11998 FT-T011999 FT-C00666 3/31/2026 23:49 Card Purchase
11999 FT-T012000 FT-C00456 3/31/2026 23:59 Cash Withdrawal

Amount_NGN Channel Device_Type Location \
0 5923.80 Mobile App Android Enugu
1 11681.12 ATM Android Kaduna
2 571.91 ATM POS Terminal Kano
3 105367.52 Mobile App Web Browser Abuja
4 3049.06 Mobile App POS Terminal Abuja
... ..
11995 9336.19 Web Web Browser Lagos
11996 16198.02 Mobile App iOS Ibadan
11997 62448.87 Mobile App ATM Terminal Lagos
11998 696.48 Mobile App Android Benin City
11999 26888.41 POS iOS Kaduna

International_Transaction Transaction_Status Risk_Review_Flag
0 No Reversed Yes
1 No Successful Yes
2 No Successful No
3 No Successful No
4 No Successful No
... ..
11995 No Successful No
11996 No Successful No
11997 No Successful No
11998 No Successful No
11999 No Successful Yes

[12000 rows x 11 columns]
```

Code

```
Fintrust.shape
(12000, 11)
```

Code

```
Fintrust.info
<bound method DataFrame.info of
Transaction_ID Customer_ID Transaction_DateTime Transaction_Type \
0 FT-T000001 FT-C01135 01/01/2026 00:00 Card Purchase
1 FT-T000002 FT-C00428 01/01/2026 00:10 Cash Withdrawal
2 FT-T000003 FT-C00469 01/01/2026 00:21 Transfer
3 FT-T000004 FT-C01015 01/01/2026 00:32 Deposit
4 FT-T000005 FT-C00870 01/01/2026 00:43 Cash Withdrawal
... ..
11995 FT-T011996 FT-C01367 3/31/2026 23:16 Card Purchase
11996 FT-T011997 FT-C00395 3/31/2026 23:27 Airtime/Data
11997 FT-T011998 FT-C00484 3/31/2026 23:38 Card Purchase
11998 FT-T011999 FT-C00666 3/31/2026 23:49 Card Purchase
11999 FT-T012000 FT-C00456 3/31/2026 23:59 Cash Withdrawal

Amount_NGN Channel Device_Type Location \
0 5923.80 Mobile App Android Enugu
1 11681.12 ATM Android Kaduna
2 571.91 ATM POS Terminal Kano
3 105367.52 Mobile App Web Browser Abuja
4 3049.06 Mobile App POS Terminal Abuja
... ..
11995 9336.19 Web Web Browser Lagos
11996 16198.02 Mobile App iOS Ibadan
11997 62448.87 Mobile App ATM Terminal Lagos
```

11998	696.48	Mobile App	Android	Benin City
11999	26888.41	POS	iOS	Kaduna

	International_Transaction	Transaction_Status	Risk_Review_Flag
0	No	Reversed	Yes
1	No	Successful	Yes
2	No	Successful	No
3	No	Successful	No
4	No	Successful	No
...	...	...	...
11995	No	Successful	No
11996	No	Successful	No
11997	No	Successful	No
11998	No	Successful	No
11999	No	Successful	Yes

[12000 rows x 11 columns]>

Code

```

Fintrust.dtypes
Transaction_ID      str
Customer_ID        str
Transaction_DateTime str
Transaction_Type    str
Amount_NGN         float64
Channel            str
Device_Type        str
Location           str
International_Transaction str
Transaction_Status  str
Risk_Review_Flag    str
dtype: object

```

Code

```

Fintrust.columns
Index(['Transaction_ID', 'Customer_ID', 'Transaction_DateTime',
      'Transaction_Type', 'Amount_NGN', 'Channel', 'Device_Type', 'Location',
      'International_Transaction', 'Transaction_Status', 'Risk_Review_Flag'],
      dtype='str')

```

Code

```

Fintrust['Transaction_Type'].value_counts()
Transaction_Type
Transfer      3549
Card Purchase 3033
Bill Payment  1475
Cash Withdrawal 1430
Deposit       1328
Airtime/Data  1185
Name: count, dtype: int64

```

Code

```

Fintrust['Device_Type'].value_counts()
Device_Type
Android      4797
iOS          2920
POS Terminal 1571
Web Browser  1447
ATM Terminal 1169
Name: count, dtype: int64

```

Code

```

Fintrust['Channel'].value_counts()
Channel
Mobile App   5102
POS          2393
Web          1869
ATM          1747
USSD         889
Name: count, dtype: int64

```

Code

```

Fintrust['Location'].value_counts()
Location
Ibadan      1550
Kano         1526
Port Harcourt 1520
Enugu        1484
Kaduna       1481
Abuja        1461
Benin City   1451
Lagos        1431
Name: count, dtype: int64

```

Code

```

((Fintrust['Risk_Review_Flag']=='Yes') & (Fintrust['Transaction_Type']=='Bill Payment')).sum()
np.int64(189)

```

Code

```

((Fintrust['Risk_Review_Flag']=='Yes') & (Fintrust['International_Transaction']=='Yes')).sum()
np.int64(177)

```

Code

```

((Fintrust['Risk_Review_Flag']=='Yes') & (Fintrust['Channel']=='Mobile App')).sum()
np.int64(990)

```

Code

```

((Fintrust['Risk_Review_Flag']=='Yes') & (Fintrust['Location']=='Lagos')).sum()
np.int64(279)

```

Code

```

((Fintrust['Risk_Review_Flag']=='Yes') & (Fintrust['Device_Type']=='iOS')).sum()
np.int64(540)

```

Code

```

pd.isnull(Fintrust).sum()
Transaction_ID      0
Customer_ID         0
Transaction_DateTime 0
Transaction_Type     0
Amount_NGN          0
Channel             0
Device_Type         96
Location            96
International_Transaction 0
Transaction_Status   0
Risk_Review_Flag     0
dtype: int64

```

Code

```

Trustfin
Customer_ID  Customer_Name  Age  Gender  City \
0  FT-C000001  Ibrahim Adeyemi  56  Male  Lagos
1  FT-C000002  Yusuf Mohammed  46  Male  Kano
2  FT-C000003  Emeka Adeyemi  32  Female  Port Harcourt
3  FT-C000004  David Williams  60  Female  Benin City
4  FT-C000005  Blessing Adeyemi  25  Male  Kaduna
...  ...  ...  ...  ...
1495  FT-C01496  Grace Adeyemi  24  Male  Lagos
1496  FT-C01497  Zainab Adebayo  26  Male  Ibadan
1497  FT-C01498  Tunde Ojo  65  Male  Enugu
1498  FT-C01499  Samuel Adebayo  25  Male  Benin City
1499  FT-C01500  Blessing Adebayo  40  Male  Lagos

Customer_Segment  Account_Type  Tenure_Months  Digital_Engagement_Score \
0  Premium  Savings  55  84.9
1  Everyday  Current  79  50.7
2  Everyday  Savings  43  56.1
3  Premium  Premium  44  54.0
4  Premium  Savings  68  85.1
...  ...  ...  ...

```

1495	Everyday	Savings	35	76.0
1496	Premium	Current	72	63.2
1497	Student	Savings	66	93.1
1498	Everyday	Premium	6	88.8
1499	Student	Current	31	51.1

	Monthly_Income_Band	Preferred_Channel	Account_Status
0	500k-999k	Mobile App	Active
1	500k-999k	Mobile App	Active
2	1m+	Mobile App	Active
3	500k-999k	Mobile App	Dormant
4	500k-999k	Mobile App	Active
...	...	...	...
1495	250k-499k	Mobile App	Active
1496	100k-249k	Mobile App	Active
1497	250k-499k	Mobile App	Active
1498	100k-249k	Mobile App	Active
1499	100k-249k	Web	Active

[1500 rows x 12 columns]

Code

```
Trustfin.shape
(1500, 12)
```

Code

```
Trustfin.describe
<bound method NDFrame.describe of
0      FT-C000001  Ibrahim Adeyemi  56  Male  Lagos
1      FT-C000002  Yusuf Mohammed  46  Male  Kano
2      FT-C000003  Emeka Adeyemi   32  Female Port Harcourt
3      FT-C000004  David Williams  60  Female Benin City
4      FT-C000005  Blessing Adeyemi  25  Male  Kaduna
...
1495    FT-C01496  Grace Adeyemi   24  Male  Lagos
1496    FT-C01497  Zainab Adebayo  26  Male  Ibadan
1497    FT-C01498  Tunde Ojo     65  Male  Enugu
1498    FT-C01499  Samuel Adebayo  25  Male  Benin City
1499    FT-C01500  Blessing Adebayo  40  Male  Lagos
```

	Customer_Segment	Account_Type	Tenure_Months	Digital_Engagement_Score
0	Premium	Savings	55	84.9
1	Everyday	Current	79	50.7
2	Everyday	Savings	43	56.1
3	Premium	Premium	44	54.0
4	Premium	Savings	68	85.1
...	...	...	...	...
1495	Everyday	Savings	35	76.0
1496	Premium	Current	72	63.2
1497	Student	Savings	66	93.1
1498	Everyday	Premium	6	88.8
1499	Student	Current	31	51.1

	Monthly_Income_Band	Preferred_Channel	Account_Status
0	500k-999k	Mobile App	Active
1	500k-999k	Mobile App	Active
2	1m+	Mobile App	Active
3	500k-999k	Mobile App	Dormant
4	500k-999k	Mobile App	Active
...	...	...	...
1495	250k-499k	Mobile App	Active
1496	100k-249k	Mobile App	Active
1497	250k-499k	Mobile App	Active
1498	100k-249k	Mobile App	Active
1499	100k-249k	Web	Active

[1500 rows x 12 columns]>

Code

```
Trustfin.dtypes
Customer_ID      str
Customer_Name    str
Age              int64
Gender           str
City            str
Customer_Segment str
Account_Type     str
Tenure_Months   int64
Digital_Engagement_Score float64
```

```

Monthly_Income_Band      str
Preferred_Channel        str
Account_Status           str
dtype: object

```

Code

```

pd.isnull(Trustfin).sum()
Customer_ID              0
Customer_Name            0
Age                      0
Gender                   0
City                     0
Customer_Segment         0
Account_Type             0
Tenure_Months            0
Digital_Engagement_Score 0
Monthly_Income_Band      0
Preferred_Channel        0
Account_Status           0
dtype: int64

```

Code

```

Trustfin.columns
Index(['Customer_ID', 'Customer_Name', 'Age', 'Gender', 'City',
      'Customer_Segment', 'Account_Type', 'Tenure_Months',
      'Digital_Engagement_Score', 'Monthly_Income_Band', 'Preferred_Channel',
      'Account_Status'],
      dtype='str')

```

Code

```

(Trustfin['Customer_Segment']).value_counts()
Customer_Segment
Everyday      711
Premium       295
Student       278
SME           216
Name: count, dtype: int64

```

Code

```

(Trustfin['Account_Type']).value_counts()
Account_Type
Savings      893
Current      378
Premium      229
Name: count, dtype: int64

```

Code

```

(Trustfin['Monthly_Income_Band']).value_counts()
Monthly_Income_Band
100k-249k      430
250k-499k      416
Below 100k     265
500k-999k     255
1m+            134
Name: count, dtype: int64

```

Code

```

(Trustfin['Preferred_Channel']).value_counts()
Preferred_Channel
Mobile App     924
Web            357
USSD           219
Name: count, dtype: int64

```

Code

```

(Trustfin['Age']).max()
np.int64(65)

```

Code

```
(Fintrust['Amount_NGN']).sum()  
np.float64(560477354.85)
```

Code

```
filt_data = Fintrust[(Fintrust['Transaction_Type'] == 'Card Purchase') & (Fintrust['Risk_Review_Flag']=='No']
```

Code

```
total_fin = filt_data['Amount_NGN'].sum()  
total_fin  
np.float64(72432592.33999999)
```

Code

```
Trustfin[Trustfin['Tenure_Months'] >= 50]  
Customer_ID Customer_Name Age Gender City Customer_Segment \  
0 FT-C00001 Ibrahim Adeyemi 56 Male Lagos Premium  
1 FT-C00002 Yusuf Mohammed 46 Male Kano Everyday  
4 FT-C00005 Blessing Adeyemi 25 Male Kaduna Premium  
10 FT-C00011 Chiamaka Bello 28 Male Abuja Premium  
12 FT-C00013 Ada Mohammed 53 Female Lagos Everyday  
... ..  
1484 FT-C01485 Mariam Ibrahim 27 Female Lagos Everyday  
1488 FT-C01489 Blessing Eze 40 Female Benin City Everyday  
1492 FT-C01493 Daniel Adebayo 24 Male Lagos SME  
1496 FT-C01497 Zainab Adebayo 26 Male Ibadan Premium  
1497 FT-C01498 Tunde Ojo 65 Male Enugu Student
```

```
Account_Type Tenure_Months Digital_Engagement_Score \  
0 Savings 55 84.9  
1 Current 79 50.7  
4 Savings 68 85.1  
10 Savings 75 70.7  
12 Savings 62 90.4  
... ..  
1484 Current 52 77.6  
1488 Current 88 48.8  
1492 Premium 78 81.4  
1496 Current 72 63.2  
1497 Savings 66 93.1
```

```
Monthly_Income_Band Preferred_Channel Account_Status  
0 500k-999k Mobile App Active  
1 500k-999k Mobile App Active  
4 500k-999k Mobile App Active  
10 Below 100k Web Dormant  
12 100k-249k Mobile App Dormant  
... ..  
1484 100k-249k Web Active  
1488 250k-499k Web Restricted  
1492 500k-999k Mobile App Active  
1496 100k-249k Mobile App Active  
1497 250k-499k Mobile App Active
```

[773 rows x 12 columns]

Code

```
Fintrust[(Fintrust['Transaction_Type'] == 'Bill Payment') ]  
Transaction_ID Customer_ID Transaction_DateTime Transaction_Type \  
12 FT-T000013 FT-C00587 01/01/2026 02:09 Bill Payment  
19 FT-T000020 FT-C01233 01/01/2026 03:25 Bill Payment  
21 FT-T000022 FT-C00555 01/01/2026 03:46 Bill Payment  
22 FT-T000023 FT-C00107 01/01/2026 03:57 Bill Payment  
27 FT-T000028 FT-C01246 01/01/2026 04:51 Bill Payment  
... ..  
11957 FT-T011958 FT-C00778 3/31/2026 16:26 Bill Payment  
11958 FT-T011959 FT-C01334 3/31/2026 16:37 Bill Payment  
11985 FT-T011986 FT-C00889 3/31/2026 21:28 Bill Payment  
11990 FT-T011991 FT-C01150 3/31/2026 22:22 Bill Payment  
11993 FT-T011994 FT-C00707 3/31/2026 22:55 Bill Payment  
... ..  
Amount_NGN Channel Device_Type Location \  
12 717.93 Web iOS Enugu  
19 523.76 Mobile App iOS Kano  
21 45562.37 POS Android Abuja  
22 1683.22 USSD Web Browser Enugu  
27 3475.92 Web POS Terminal Ibadan
```

...	...	...	...	...
11957	7755.26	ATM	ATM Terminal	Port Harcourt
11958	22779.06	POS	POS Terminal	Kano
11985	8064.06	Web	POS Terminal	Abuja
11990	73992.29	POS	ATM Terminal	Kaduna
11993	7767.84	Mobile App	iOS	Kano

	International_Transaction	Transaction_Status	Risk_Review_Flag
12	No	Successful	No
19	No	Successful	No
21	No	Successful	No
22	No	Successful	No
27	No	Successful	No
...	...	...	...
11957	No	Successful	No
11958	No	Successful	No
11985	No	Successful	No
11990	No	Successful	Yes
11993	Yes	Successful	No

[1475 rows x 11 columns]